We are at the end of our excursus on practical ways to use Bloom filters; it’s worth mentioning that other applications of Bloom filters include rate limiters, black lists, synchronization speedup, estimating the size of joins in DBs.

4.8 Why Bloom Filters Work[15]

So far, we have asked you to take for granted that Bloom filters do work as we described. Now it’s time to look more closely and explain why a Bloom filter actually works. Although this section is not strictly needed to implement or use Bloom filters, reading it might help you understand this data structure more in depth.